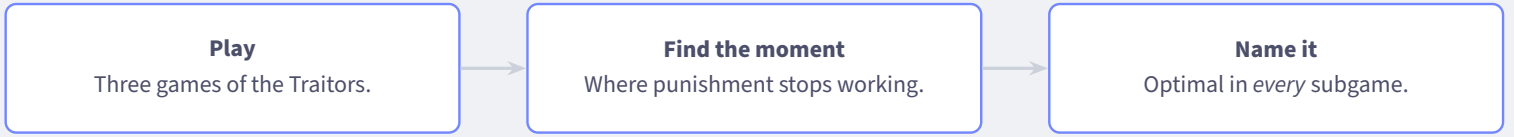
10

Subgame perfection

A threat that stops being worth carrying out

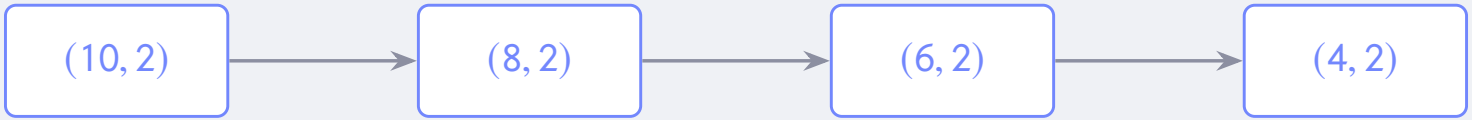


Each round has two phases.

Night The Traitors agree silently on one Faithful to remove. Everyone else has their eyes shut.

Day Everyone votes. The most-voted player is banished and their role is revealed. Ties are broken at random.

So every round costs two players: one banished, one murdered.



Writing (n, m) for n players and m Traitors: the Traitors win as soon as $2m \geq n$, the Faithful once every Traitor has been banished.

1. Everyone is dealt a role, face down. Look at it. Do not show it to anybody.
2. The Traitors learn who the other Traitors are. The Faithful learn nothing.
3. Sit in a circle and stay in that order for the whole game. Your position matters later.

Today: _____ players, of whom _____ are Traitors.

Everything a Faithful knows comes from how people vote. That is the whole game.

1. Play the two phases as described. Vote however you like.
2. Nobody is telling you how to vote. Talk, argue, accuse.
3. We stop after a round or two. The point is not to finish.

The question to hold on to:

if the Traitors are working together, can you tell?

New roles, fresh game, and now a rule everybody follows.

Rule	Vote for the next surviving player to your left.
Why	Under the rule everyone receives exactly one vote, so the banishment is uniform: the same $1/n$ chance as voting at random. It costs the Faithful nothing.
Catch	Break the rule and it shows. One player gets two votes and somebody gets none, and everybody can see who did it.
Punishment	Anyone seen to break the rule is banished next.

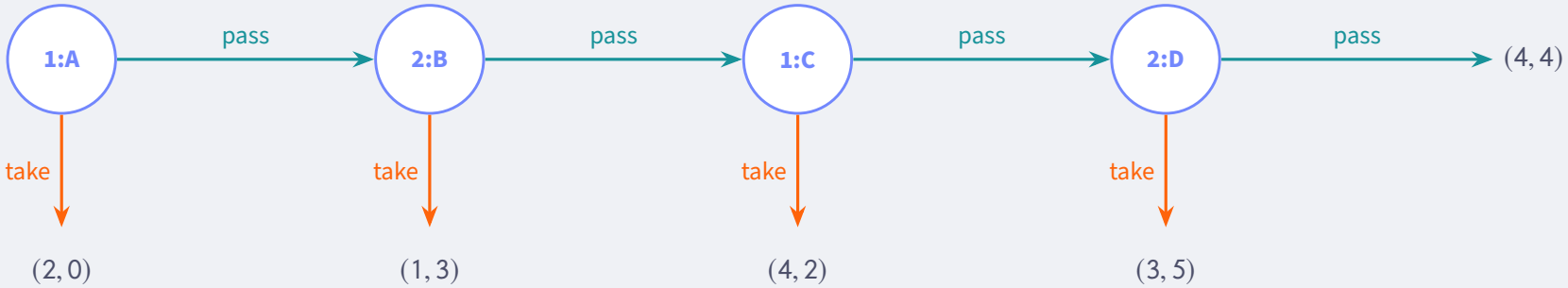
Play this one on towards the end. Watch what happens as the circle gets small.

Same rule, same punishment, with one change.

New Anyone may break Vote-Left now and then, for no reason at all. No warning, no announcement.

The question to hold on to:

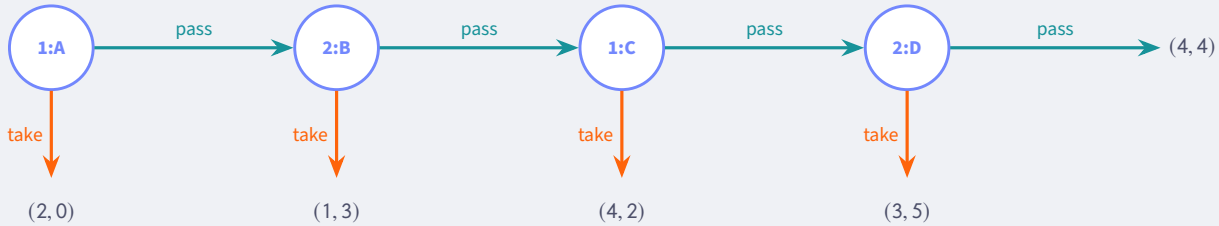
now that an odd vote might just be somebody playing around, is the threat to banish a suspect still worth carrying out?



Payoffs are (player 1, player 2). Every node carries the player who moves there and a letter: player 1 moves at A and C, player 2 at B and D.

Pick a player. Write a strategy for them on paper, then swap and write one for the other.

1. A strategy gives an action at *every* node where that player moves, so the table has one row per node.
2. Fill in both rows, even if the first action ends the game. A strategy says what you *would* do, not only what happens.



Player 1

Node	Action
1:A	
1:C	

Player 2

Node	Action
2:B	
2:D	

Pass at A with *take* at C means: pass at the first circle, and take at the third if we ever get there.

Two nodes, two actions at each, so each player has $2 \times 2 = 4$ strategies. List all four.

Keep the same order as everybody else in the room: it is the order the rows and columns of the grid will take.

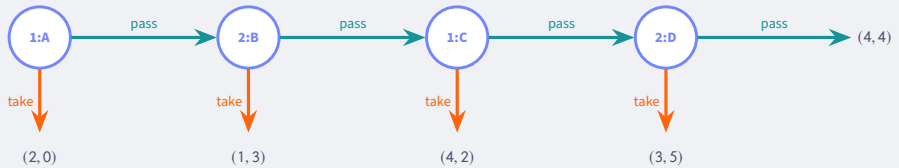
Player 1

	A	C
1		
2		
3		
4		

Player 2

	B	D
1		
2		
3		
4		

Rows are player 1's strategies, columns are player 2's, in both matrices. Fill a cell by tracing the path that pair of strategies produces.



A: what player 1 gets

B: what player 2 gets

The pair (A, B) is an ordinary normal form game, so we find its equilibria the way we always do.

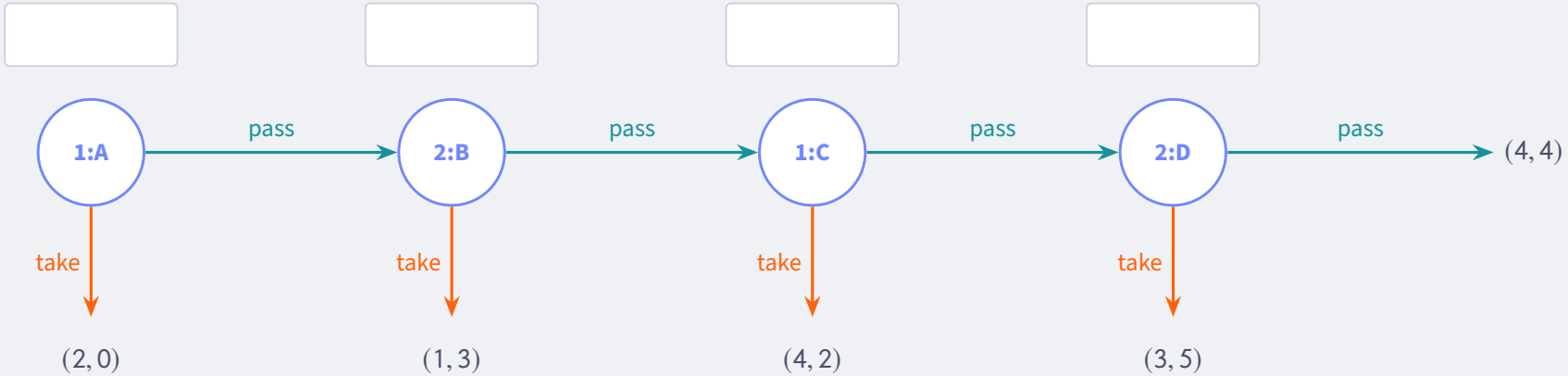
1. Down each column of A, underline player 1's best payoff.
2. Across each row of B, underline player 2's best payoff.
3. A position underlined in both matrices is a pure Nash equilibrium. There is more than one.

A

B

Only one of those equilibria survives backward induction. The others rest on a threat nobody would carry out.

1. Start at D, the last node, and mark the action player 2 prefers there.
2. Work leftwards: at each node, compare the payoffs the marks to its right lead to.
3. Read off one action per node. That pair of strategies is the subgame perfect equilibrium.



What we just did in the room is **Question 1** on the Subgame Perfection page.

(a)	Draw the extensive form.	[3]
(b)	Backward induction: the subgame perfect equilibrium and its payoffs.	[5]
(c)	Show (comply, banish) is a Nash equilibrium.	[5]
(d)	Explain why it is not subgame perfect, and how that mirrors the class game.	[6]
(e)	For general q, find where the threat deters and where it is credible.	[4]

vknight.org/gt/topics/subgame-perfection.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Roles, and who can see what	**2.2.2**	Games: Definition: Information Set
Vote-Left as an instruction	**2.2.5**	Games: Definition: Strategy in Extensive Form Games
Writing the centipede down	**2.2.1**	Games: Definition: Extensive Form Game
Four strategies each	**2.2.6**	Games: Mapping Extensive Form Games to Normal Form
The two matrices	**2.2.3**	Games: Definition: Normal Form Game
Underlining best responses	**5.2.1**	Nash Equilibrium: Definition: Nash Equilibria
The moment the threat died	**6.2.1**	Subgame Perfection: Definition: Sequential rationality
Working back from the last node	**6.2.2, 6.2.4**	Subgame Perfection: Backward induction, Subgame
The equilibrium that survived	**6.2.5**	Subgame Perfection: Definition: Subgame perfect equilibrium
Examinable	**6.2.3**	Subgame Perfection: Theorem of existence of Nash equilibrium in games of perfect information, and proof

vknight.org/gtb/chapters/games, vknight.org/gtb/chapters/nash-equilibrium, vknight.org/gtb/chapters/subgame-perfection

Quick quiz before you go: vknight.org/gt/topics/subgame-perfection.html

Before the next class: read the Subgame Perfection chapter as far as the subgame perfect equilibrium, and attempt Question 1.